Predictive Model for High Energy Consumption Processes in the Steel Manufacturing Industry using Machine Learning Tools: Industry 4.0 Pathway

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***Abstract*—The iron and steel industry is responsible for 7% of the total emissions and 11% of carbon emissions at a global level. Industry 4.0 is one of the triggered topics for the decarbonization of the steel industry. Industry 4.0 emphasizes automation, process optimization, and the use of AI and ML for decision-making. The study's approach is methodological, where we started with energy audits carried out in three steel mills to pinpoint how machine learning (ML) techniques can be effectively used in search of relevant energy efficiency (EE) benchmarks. The energy audits were carried out from visits to steel mills, collection of production and energy data, categorization of processes, identification of energy-intensive areas, setting up possible key performance indicators (EnPIs) at the process and mill levels which paves the way for the energy saving opportunities. This study gives an insight into the challenges in the implementation of industry 4.0 pathway in the steel industry, using an application available in MATLAB i.e., regression learner. The authors discuss the current systems, processes, and practices in steel mills, identifying gaps e.g., the type and importance of the data to be collected at each energy intensive process, that need to be addressed for the implementation of Industry 4.0, with a particular focus on the use of ML and AI. This work primarily contributes to advancing the application of ML within the steel industry.**

***Keywords*—decarbonization, steel, industry, EnPI, machine learning, energy efficiency, industry 4.0**

I. Introduction

According to International Energy Agency the direct emissions from the production of steel are between 1.4 and 1.85 tonnes of CO_2 per tonne of steel produced [1]. The steel sector is not on track to meet the net zero requirements [2], with its emissions still increasing. Hence, there is an urgent call for increasing the energy efficiency (EE) in this sector, and machine learning (ML) techniques for intelligent data analysis [3]. The authors of [4] focused on the role of ML for a number of predictive maintenance tasks, where data is taken from the very common sensors installations around a plant. Specifically, monitoring key performance indicators (EnPIs) is one of the methods for improved energy efficiency of steel industry, in [5] the authors have used a regression model for optimizing production process in small and medium sized enterprises. Furthermore, in [6] the authors have developed the deep learning predictions of the mechanical properties of the steel. They have developed a model for the prediction of different process parameters i.e. temperature, dimension and speed for the hot rolling process. In another study, a predictive model by regression is being presented for energy consumption in the steel industry. The considered inputs are: kWh, kVarh, power factor both leading and lagging, emissions and working hours [7]. It gives an insight of how the process level EnPIs are important and how ML can be a part of an energy intensive sector.

This work aims to investigate the methodology at the sector-specific level for the potential energy saving opportunities by using ML techniques for steel production, with a first example of application. The work is broken into the following research questions.

- Why it is important to collect the data in steel production at different process level and relate them?
- What type of data is important to collect from an energy manager standpoint?
- How ML can be effective in analyzing the data and setting up energy performance indicators?

The paper is structured as below. First of all, the high energy consumption processes are identified based on the data from the case studies (3 steel mills). Second, data sets for energy intensive processes are prepared by taking into account the allegedly parameters, important to monitor, which will likely effect the quality, the chemical composition of steel and the energy consumption. Third, how regression learner, which is a ML tool, efficient in identifying relationships between input and outcomes, is helpful on predicting the EE of a process. Finally, the discussion and conclusion are presented.

II. Methodology for energy performance indicators and ML

This study has been carried out by involving three steel mills. The effort is to analyze the the high energy consumption activities in steel mills with the help of ML tool. A significant time has been spent in the steel mills for understanding the whole process, collecting data of different activities, building an energy model, identifying EnPIs and finding out the the possible interventions. The authors reported the whole methodology in detail in [8]. Steel manufacturing process (SMP) refers

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to the succession of processes that converts the raw material into different products of steel like bars, coils, ingots etc. The whole SMP is categorized into different sub-processes: (i) steel making , (ii) casting and conditioning, (iii) forging, (iv) hot rolling, (v) heat treatment, (vi) surface treatment and finishing (straightening, cold drawing, peeling, grinding and testing). Each process is categorized further into three sets of activities: (a) principle (responsible for production), (b) auxiliary (e.g. compression, lightening, air conditioning..) and (c) general activities (e.g. canteen, quality control, lightening..). Also, all the relevant actions/items, involved in each activity , have been identified to establish the methodological set up.
Data of energy/resources consumption (electricity, fossil fuels, water, ...) and production (different forms of steel e.g. crude steel, remelted ingots, forged ingots, coils and bars) has been collected and analyzed for different processes in SMP. Each process has different equipment and each equipment is further categorized into three sets of activities (a, b, and c as reported above). The specific (rated over t of steel) energy consumption (kWh/t electricity, GJ/t overall energy, Sm^3/t) , price (Euro /t) and carbon emissions (t CO_2/t) of different processes along with the production of steel can be spotted from Fig. 1. Data is the average value of mill 1, mill 2 and mill 3. This specific figure reports an electric arc furnace based steel mill which is two of the cases studied, hence high electric energy is consumed in the steel-making process. Here, steel making refers to charging, melting and elaboration, which encompasses different processes like electric arc furnace (EAF), vacuum arc remelting (VAR), electroslag remelting (ESR), argon oxygen decarburization (AOD).

In the next subsections, we will report the data sets observed to optimize the energy consumption, chemical composition of the steel, purity level of ingots for EAF, VAR, ESR, AOD and air compression system and implementation of regression learner that is a ML tool. These processes are selected based on the high energy consumption as reported in Fig. 1.

A. Electric arc furnace

EAF is considered as one of the highest energy consumption systems and can be used to produce a wide range of steel. The EAF process involves charging the furnace with raw material, melting it by using the electric arc, refining it by adding different chemicals and tapping. The quality of the steel after processing in EAF is very important to monitor. According to [9] the average energy consumption in the EAF ranges from 400 to 500 kWh/ton, hence, the control of such indicator is very important in the EAF. In [10], a study related to 'energy consumption modeling in EAF' is presented and the following parameters have been identified being able to affect consumption: scrape, additives, heat number and steel grade. According to [11] the desired composition of stainless steel at the end of EAF is carbon (0.8-1.65%), chromium (14.83-20.50 %), nickle (5.83-9.89%), silicon (0.03-0.68%) and iron (69.09-76.34%) with the electricity range of 419-601 kWh/t.

In another study , different parameters scrape weight, tapping weight, slag, temperatures, power on/off time, added gas and oxygen and energy loss are considered for the energy management in EAF [12].

The parameters, required to be regularly monitored for optimizing the process based on the literature available and study carried out in steel mills, are reported below:

DATA SET: Parameters for optimization of energy consumption and quality of steel for EAF

Input: Composition of the scrape: % of iron, carbon and manganese
Input: Electrode power in kWh
Input: The rate of oxygen injection in Nm^3/h
Input: Addition of fluxes in kg
Input: Charging time in h, min
Input: Temperature before melting in °C
Input: Temperature of molten steel in °C
Input: Composition of the final steel in %

Output: Quality of steel based on tensile strength, hardness in MPa
Output: Energy consumption in kWh

The authors of [13] discussed the energy consumption in EAF as a function of oxygen consumption. They found out that efficient consumption of oxygen will lead to less energy consumption. They found an oxygen electrical energy equivalence as 3.6 kWh/t. It has been reported that the kWh to raise the temperature of 1 t of steel is 1540 °C to 1600 °C.

B. Vacuum Arc Remelting and ElectroSlag Remelting

VAR and ESR are the processes in secondary metal refining, one of the primitive casting techniques. Both processes produce steel ingots and nickel alloys. In VAR, the metal to be refined is used as electrode and an electric arc is responsible to melt this electrode. The molten metal dips and takes the shape of ingots. This whole process is carried out in vacuum chamber to avoid atmospheric contamination and reduce the oxides. In ESR, the electrode is dipped into the molten slag which is also conductive. An electric current passing is responsible for melting of metal [14]. Control of inclusion of non metallic materials or oxide cleanliness is one of the keys to get high grade steel [15]. It can be done by continuous monitoring of VAR and ESR outputs by using the predictive models reported in section III. The parameters for the optimization are below.

DATA SET: Parameters for quality of steel and energy consumption for VAR and ESR

Input: Electrode composition % of carbon, chromium and nickle
Input: Electric arc current in A
Input: Level of vacuum in the chamber in $Torr$ (for VAR only)
Input: Duration of melting process in $hours$
Input: Rate at which the ingots cooled in °C$/min$

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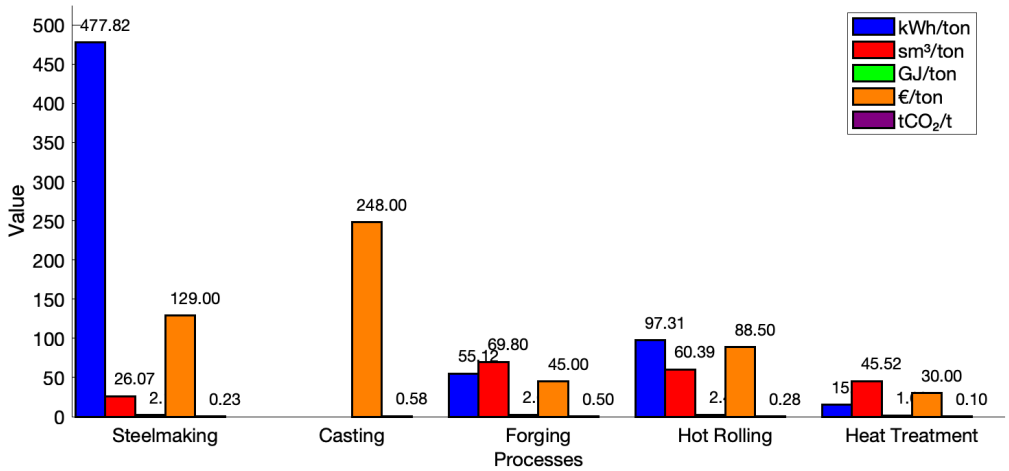

Fig. 1: Specific energy indicators (values rated over the t of steel) for relevant processes: average value for the 3 studied steel mills

Input: Temperature of the slag in °C (for ESR only)
Input: Composition of the slag in % (for ESR only)

Output: Quality of steel based on tensile strength, hardness in MPa
Output: Total energy consumption in GJ
Output: Chemical composition of final ingots in %
Output: Purity level of the final ingots in ppm

C. Argon Oxygen Decarburization

AOD is a refining process to get desired steel through addition of different materials at different stages. The stages involved are: blowing, decarburization, alloying and tapping. Argon and oxygen are blown through lance into the molten steel. Carbon reacts with oxygen to form CO_2 and CO in decarburization and been removed from the steel. Additional alloying elements like nickle and chromium are added to the desired composition [16]. Parameters to be monitored for this process are reported below.

D. Argon Oxygen Decarburization

DATA SET: Parameters for refined steel after AOD

Input: Composition of molten/scrape steel in %
Input: Flow rate of argon in Nm^3/min
Input: Flow rate of oxygen in Nm^3/min
Input: Duration of melting process in $hours$
Input: Temperature of the molten steel in °C
Input: % of chromium added
Input: % of nickle added
Input: Total energy consumption in AOD in kWh

Output: % of final steel composition

E. Air compression system

Auxiliary services also consume a significant amount of the energy in steel mills. The authors of [17] discussed the potential energy saving opportunities in compressed air system, which took our interest towards the auxiliary systems. The other auxiliary systems which consumes a significant amount of energy are: pumping system, purification system, ventilation system, water cooling system, lightening system, local air conditioning system. One of the auxiliary services, considered here in detail for setting up EnPI, is air compression systems. Air compression system uses $4-6\%$ of the electricity consumed in whole steel mill. According to the [18] these systems comprises of $5-20\%$ of the annual electricity cost. Parameters which can be effective in considering the air compression quality are below.

DATA SET: Parameters for optimization of EE of compressors

Input: Load utilization factor hours/year
Input: Volume of the gas flow in $Nm^3/year$
Input: Idle working hours of compressors
Input: Average electric power absorbed when idle in kW
Input: Average electric power absorbed under load in kW
Input: Relative pressure in MPa

Output: Energy efficiency of compressors in $kWh/Nm^3/ton$

III. Methodological Data Analysis Framework for Steel Mill

In the field of ML for understanding complex relationships between variables and predicting outcomes, regression analysis has emerged as a first powerful and straightforward tool. In the context of steel manufacturing, this technique has proved to

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serve as significant promise in optimizing power consumption and improving overall process efficiency. Regression models excel at identifying relationships, making them particularly useful in complex industrial settings, multivariate regression techniques can effectively capture the complex relations between various process parameters in steel production. The steel industry faces ongoing challenges in optimizing energy consumption, product quality, and process efficiency. The data provided on EAF, VAR, ESR, AOD and air compression system presents an opportunity to leverage ML regression analysis for process improvement [20].

A. Data Acquisition and Pre-processing

The initial phase involves data collection, data cleaning and data normalization from various steel production processes. Process parameters and output metrics are systematically gathered from each production stage in data collection. Raw data undergo pre-processing to address missing values, outliers, and inconsistencies n data cleaning. Input features are scaled to a common range, ensuring equal weighting in the subsequent learning process in data normalization.

B. Model Selection and Training

Steel production optimization is formulated as supervised learning problem with multiple inputs and outputs. Several models are evaluated in which multiple linear regression (MLR) assumes a linear relationship between inputs and outputs, making it simple but limited for complex data. Random Forest combines multiple decision trees to improve accuracy and handle non-linear relationships. Gradient boosting machines (GBMs) build trees sequentially, each correcting errors from previous ones, making them powerful but prone to over fitting if not tuned well. Neural Networks use interconnected layers of nodes to model complex patterns, offering high flexibility but requiring large datasets and computational resources [19]–[21]. They're highly flexible and can capture intricate patterns in data predicting EE in steel production.

C. Model Evaluation

A comprehensive evaluation strategy is implemented to assess model performance, hence models are evaluated using:

- Relative Root Mean Squared Error (rRMSE)
- R-squared (R^2)
- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)

Analysis of feature importance is conducted to understand key drivers of steel quality and energy consumption. Examination of prediction errors is performed to identify potential model weaknesses or data anomalies.

D. Interpretability and Explainability

To ensure transparency and foster trust in the model's predictions, several interpretability techniques are employed [22]. This methodological model provides a framework for applying ML techniques to steel production optimization, encompassing the entire process from data acquisition to model deployment and ongoing monitoring, with a strong emphasis on interpretability and ethical considerations.

E. Performing ML on available data sets on the case study

To analyze and predict the EE of compressors i.e. the energy consumed by the air compressors facility of the case study (3 steel mills) to deliver $1Nm^3$ for each t of produced steel will be considered as the predicted value to look for when 12 variables are the input as shown in Fig. 2. Thus, a ML approach is tested as illustrated in Fig 3. This workflow allows for systematic development, tuning, and comparison of *multiple* ML models to find the best performing one for the given data and predicting the EnPI of compressors. The process involves reading input and output data, splitting it into training and test sets, and then iterating through model selection, parameter tuning, cross-validation, and training. Once a model is trained, it is used for prediction on the test set, and errors are calculated. This process is repeated for all selected models before achieving the final results. This approach can thus be applied to processes, services and even to the whole SMP.

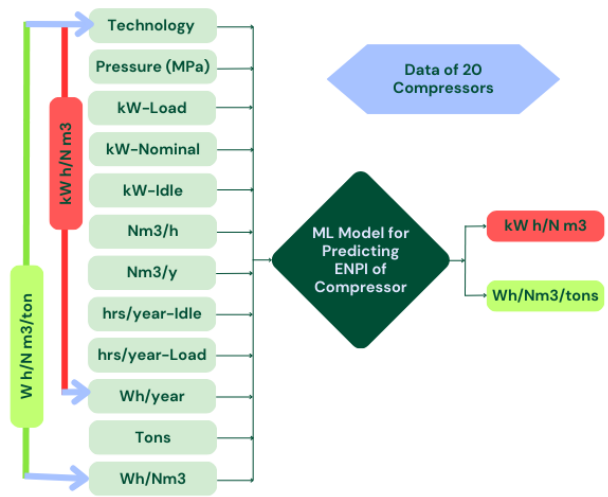

Fig. 2: Machine learning model overview for two EnPIs

Energy efficiency of compressors: in terms of electricity consumption, the air compressors facilities are reckoned to be critical loads, being a service in almost all the departments of a SMP. From this knowledge, it is thus easy to upgrade the compressors and make/calculate the significant energy and economic benefits of any relevant intervention. As explained above, the energy efficiency of the compressors refers to efficient compression and it is measured by kWh/Nm^3. It is a well known metric, which has been always widely used for such a facility, across various manufacturing sectors, to check the efficiency running of such machines.

A lower kWh/Nm^3 0.1-0.15 [23] reflects a more efficient compression which is a result of less energy use and cost savings. It also indicates less direct emissions. Monitoring this parameter is also helpful in predictive maintenance if data for instance are available on a daily basis. The index can be

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computed by rating the consumed energy on a period T over the air compressed volume on the same period T. Even if the energy consumed can be assessed by knowing the way of operation of the machines -both fixed speed and variable speed- (mainly: the n. of hours in load and unload mode, the n. of cycling between the two modes, idling time from unload to off mode) nevertheless there are other parameters (such as the temperature, the humidity, the dirtiness of air, the state of the pipelines, the configuration of the network, ...and many other conditions related to the manufacturing process) which heavily affect the energy and volume computation.

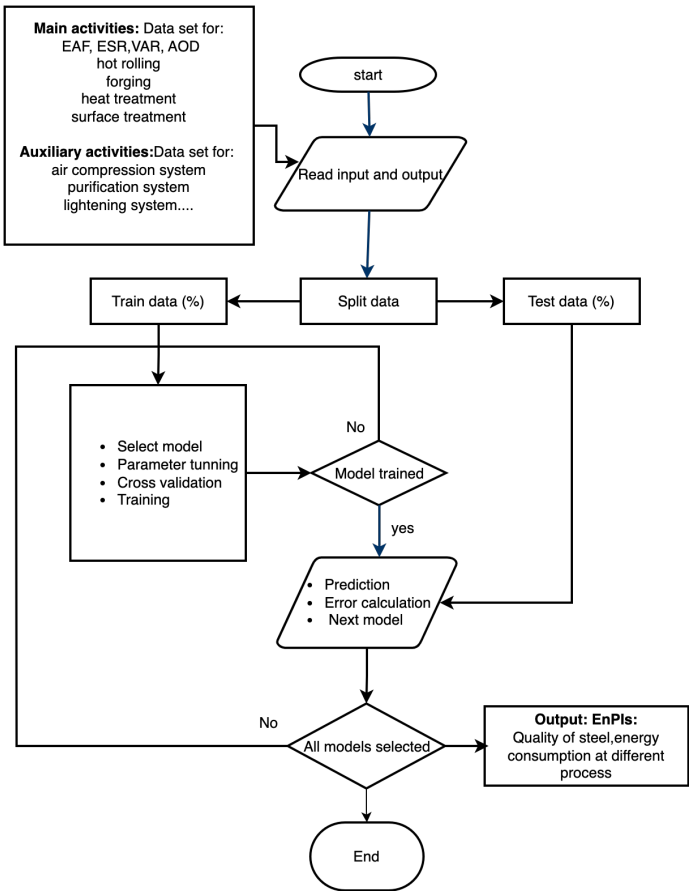

Fig. 3: Machine learning systematic approach for model evaluation

Once period T of investigation is chosen, in this case one year, the participating factors for EnPIs are chosen as shown in Fig. 2: For proposed case study total number of participating compressors are 20. The real time value are trained by using the regression learner and results are shown in Fig. 4.

The training result shows a relative root mean square error (rRMSE), R-Squared value, MSE, and MAE. Specifically: lesser rRMSE means the better prediction results; R-Squared shows the improvement in prediction after using the regression model as compared to the mean model. This first attempt can of course be improved, but a better granularity should be used. So the question is: with the current granularity how can we improve the prediction? A new EnPI, using pretty much

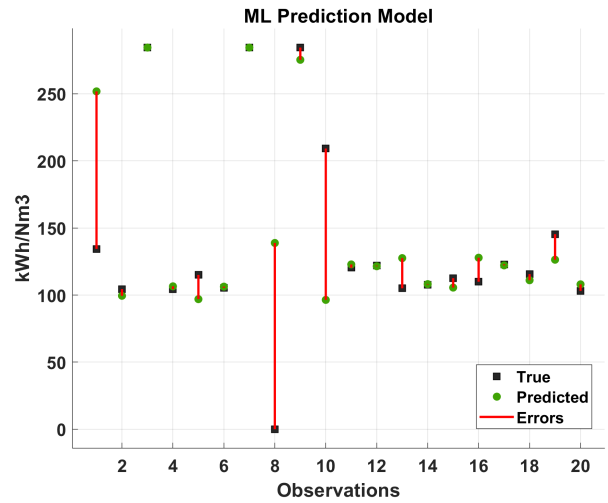

Fig. 4: Prediction vs. real kWh/Nm^3 using toolbox

the same input parameters, but including information on the process, can thus be the answer. This trial is performed below.

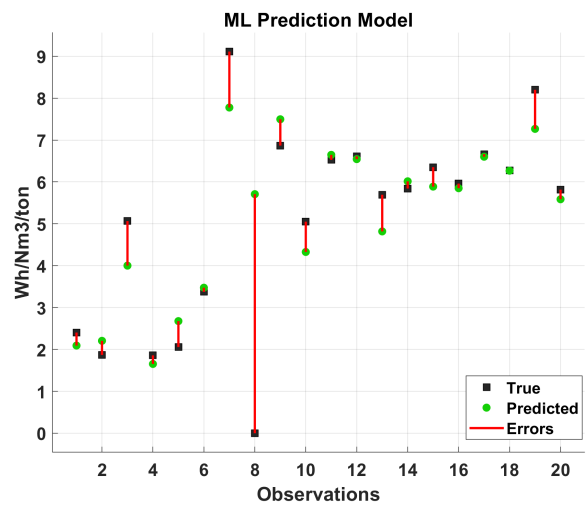

Fig. 5: Prediction vs. real '$Wh/Nm^3/ton$' by using toolbox

A new alternative metric ($Wh/Nm^3/ton$) is thus checked for better predictive results. To measure this EnPI, the areas of compressors installed are identified, for instance 3 compressors were installed in steel making, 5 in hot rolling and 2 in forging. The results are shown in Fig. 5. A comparison of metrics of both EnPIs are presented in Table I.

TABLE I: Comparison of regression models for both EnPIs

Metric	kWh/Nm^3(EnPI$_1$)	$Wh/Nm^3/ton$ (EnPI$_2$)
rRMSE	34.93	27.5
R-Squared	0.55	0.71
MSE	2376.0	1.9515
MAE	24.216	0.70339

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The comparison of regression models reveals that model of $EnPI_2$ outperforms model of $EnPI_1$ in several key metrics, including lower rRMSE, higher R-Squared, lower MSE, and lower MAE, indicating greater accuracy and better fit. The rRMSE and R^2 is a dimensionless metric allowing for a fair comparison between the two EnPI models. The superior rRMSE and R-squared metrics for the $EnPI_2$ model demonstrate its better overall predictive performance compared to the $EnPI_1$ model, despite the different units of the target variables.

IV. Discussion and Conclusion

The contribution of this work is focused on how to support the use of ML in the steel industry. The case study considers data from three steel mills providing a preliminary perspective on energy consumption patterns. The identification of high-energy-consuming processes (EAF, VAR, ESR, AOD, and air compression systems) allowed for a focused approach to energy optimization. The study shows how the choice of the most suitable indicator (EnPI), to be predicted (Wh/(Nm^3/ton) vs. the most common kWh/Nm^3) in case of limited -or first attempts in the use of ML- data, can provide more insightful and precise information. By analyzing the model's performance a strong predictive power on the chosen EnPI is shown. In the short-term, more improvements are expected by a better fine-tuning: for instance, selecting more location-specific data, with higher temporal granularity (a week, a month, every change in the batch process, ...) and a more dedicated data gathering on the production itself (sub-process level) are likely to guarantee the setup and use of a robust and rigorous procedure, with higher and higher converging results.

This preliminary study strengthens the transformative potential of Industry 4.0 technologies, particularly the ML, in supporting the progresses towards decarbonization of the the steel manufacturing. By applying data-driven approaches to energy efficiency (thus emission reduction), the research provides an additional suggestion for integrating advanced analytics into traditional steel production processes, shifting operational decision-making from experience-based to data-informed strategies. Ultimately, this research lays the groundwork for a paradigm shift towards data-driven, energy-efficient, and sustainable steel production, with far-reaching implications for reducing energy consumption, lowering carbon emissions, and enhancing operational efficiency in the face of environmental pressures and competitive challenges.

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